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University Examinations 2016/2017

FIRST YEAR FIRST SEMESTER EXAMINATION
FOR THE DEGREE OF
MASTER OF SCIENCE IN APPLIED STATISTICS

SMS 5126: MULTIVARIATE ANALYSIS

DATE: DECEMBER 2016 **TIME: 3 HOURS**

INSTRUCTIONS: *Answer question **one** and any other **two** questions.*

QUESTION ONE (30 MARKS)

a) Highlight areas where multivariate analysis has been applied extensively (6 marks)

b) Consider two variables j and k with covariance σjk. Construct the population correlation (3 marks)

c) Given $\sum = \begin{bmatrix} 64 & 2 & 4 \\ 2 & 16 & 3 \\ 4 & 3 & 9 \end{bmatrix}$ find the correlation matrix.

d)

(i) Let $Y = \underset{\sim}{a}^T \underset{\sim}{x}$

Where $\underset{\sim}{x}$ is a random vector, $\underset{\sim}{a}$ is a p-dimensional vector of constants.

Determine the mean and variance of y. (4 marks)

(ii) Let $x \sim N_3\left(\underset{\sim}{\mu}, \Sigma\right)$, where $\underset{\sim}{\mu} = \left(2\ 0\ 5\right)$

and $\sum = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 4 \end{bmatrix}$ suppose

$y_1 = x_1 + x_2 + x_3$ and $y_2 = x_2 - 2x_3$

Determine the mean and covariance of $\underset{\sim}{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ (6 marks)

e) Show that

(i) $\underset{\sim}{\bar{x}}$ is an unbiased estimator for μ and has covariance matrix $\frac{1}{n}\Sigma$ (5 marks)

(ii) $\frac{n}{n-1}s_n$ is an unbiased estimator for Σ (6 marks)

QUESTION TWO (20 MARKS)

a) Discuss the objectives of multivariate methods (10 marks)

b) Most multivariate methods are based upon the simple concept of distance. For two points say P and Q, discuss several proposals for defining distance and select a statistically relevant alternative (10 marks)

QUESTION THREE (20 MARKS)

a) Discuss the geometry and random sampling for sample mean, variance and correlation (4 marks)

b) If $x \sim \mu_P\left(\underset{\sim}{\mu}, \Sigma\right)$ and Σ is positive definite show that

$\left(\underset{-}{x} - \underset{\sim}{\mu}_p\right)\Sigma^{-1}(x-\mu) \sim \chi_p{}^2$ (6 marks)

QUESTION FOUR (20 MARKS)

Let $\begin{pmatrix} \underset{\sim 1}{x} \\ \underset{\sim 2}{x} \end{pmatrix} \sim \mu_p\left[\begin{pmatrix} \underset{\sim 1}{\mu} \\ \underset{\sim 2}{\mu} \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}\right]$

Then $\underset{\sim 1}{x} \Big| \underset{\sim 2}{x} = x_2 \sim \mu_q\left(\mu_{21}, \Sigma_{12}\right)$

Show that

$$\mu_{1.2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}\left(\underset{\sim}{x}_2 - \underset{\sim}{\mu}_2\right)$$

$$\Sigma_{1.2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Obtain the covariance of the residual and fitted set.

QUESTION FIVE (20 MARKS)

a) Given $x \sim \mu_4\left[\begin{pmatrix}1\\2\\0\\4\end{pmatrix}, \begin{pmatrix}4 & 0 & 1 & 3\\0 & 4 & 1 & 3\\1 & 1 & 3 & 1\\3 & 1 & 1 & 9\end{pmatrix}\right]$

Find the conditional correlation of

$\begin{pmatrix}x_1\\x_3\end{pmatrix}$ given $X_2 = x_2,\ \ X_4 = x_4$

Obtain the conditional covariance matrix.

Compute partial correlation between x_1 and x_3 with controlling the values of x_2 and x_4

Compute correlation coefficient between x_1 and x_2 (15 marks)

b) If $x_1 \ .\ .\ x_n$ is a random sample from a population with mean vector

$\underset{\sim}{\mu} = \begin{bmatrix}\mu_1\\ \mu_p\end{bmatrix}$

For any $\varepsilon > 0$, $prob\left\|\bar{x} - \underset{\sim}{\mu}\right\| > \varepsilon) \to 0$ as $n \to \infty$. Show that s is a consistent estimator for Σ i.e.

For $\varepsilon > 0$ $p\left[\left|s_{ij} - \sigma_{ij}\right| > 0\right] \to 0$ as $n \to \infty$ $s_{ij} \overset{p}{\to} \sigma_{ij}$ (5 marks)